

Local Nonlinear Projection-Based Reduced-Order Models

Sebastian Ares de Parga³, Radek Tezaur¹, Charbel Farhat^{1,2}

¹Department of Aeronautics and Astronautics

²Institute for Computational and Mathematical Engineering
Stanford University, Stanford, CA 94305, USA

³CIMNE, Barcelona 08034, Spain

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Problem: 2D Burgers, mesh 250×250 .

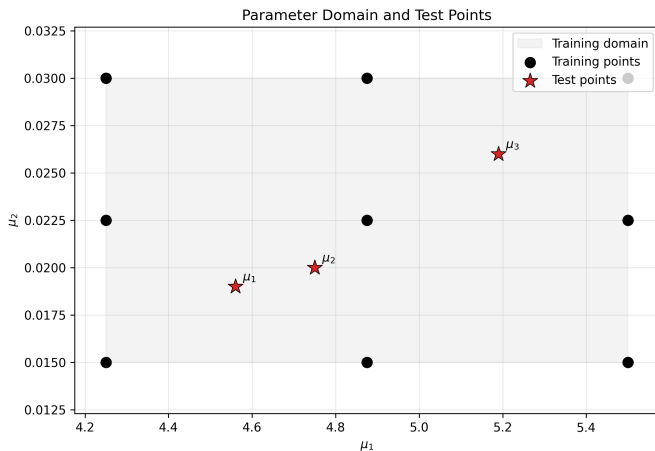
Test points:

$$\mu_1 = (4.56, 0.019), \quad \mu_2 = (4.75, 0.020), \quad \mu_3 = (5.19, 0.026).$$

Methods included:

- ▶ Global HPROM (linear POD)
- ▶ Global HQPROM (quadratic manifold)
- ▶ Global HPROM-GPR
- ▶ Local HPROM
- ▶ Local HQPROM
- ▶ Local HPROM-GPR

Parameter Domain and Test Points



Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^{\star} = \arg \min_{\mathbf{q}} \frac{1}{2} \|\mathbf{r}(\mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}))\|_2^2.$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta \mathbf{q}$ is obtained from

$$(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi}) \Delta \mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}}, \quad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \quad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta \mathbf{q}.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}, \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad n = 96.$$

Hence

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \quad \boldsymbol{\psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}), \quad n = 39.$$

Thus

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}[\mathbf{H}(\mathbf{q} \otimes \mathbf{q})].$$

Split basis and reduced coordinates:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{q}_{\text{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\text{GPR}}(\mathbf{q}), \quad n = 20, \bar{n} = 131.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}),$$

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}} = \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}_{\text{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}_{\text{GPR}}}{\partial \mathbf{q}}.$$

Local ROMs: Piecewise Manifolds

Partition the training snapshots into clusters $k = 1, \dots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q},$$

$$\Phi_k^{\text{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\text{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\text{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^+ = \arg \min_{\ell=1,\dots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \quad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold $\Phi_{k^-}^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$:

$$k^+ = \arg \min_{\ell=1,\dots,N_c} S_{k^-, \ell}^{\text{lin}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{lin}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + d_{k^-, \ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Cluster Selection: Final Form (HQPROM and HPROM-GPR)

HQPROM selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{quad}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{quad}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) + d_{k^-, \ell},$$

$$\mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell})^T \mathbf{H}_{k^-} (\mathbf{q} \otimes \mathbf{q}).$$

HPROM-GPR selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \bar{\mathbf{g}}_{k^-, \ell}^T \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}) + d_{k^-, \ell}.$$

$$\bar{\mathbf{q}}_{k^-} := \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}_{k^-}), \quad \bar{\mathbf{g}}_{k^-, \ell} := \bar{\mathbf{V}}_{k^-}^T (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell}).$$

Cluster Change: Linear and Nonlinear Transfers

$k^+ = k^- \Rightarrow$ no transfer, $k^+ \neq k^- \Rightarrow$ solve transfer problem.

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

Linear local target

$$\mathbf{q}_{k^+}^{\text{lin}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - (\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+} \mathbf{q}) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\text{lin}} = \mathbf{V}_{k^+}^T (\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

Nonlinear local targets (HQPROM / HPROM-GPR)

$$\mathbf{q}_{k^+}^{\text{nl}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\text{nl}}(\mathbf{q}) \right\|_2^2.$$

Cluster Change: Nonlinear Least Squares

For nonlinear local manifolds, define

$$\mathbf{r}_{\text{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \quad \mathbf{J}_{\Phi}(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

Gauss–Newton at iterate $\mathbf{q}^{(j)}$:

$$(\mathbf{J}_{\Phi}^T \mathbf{J}_{\Phi}) \Delta \mathbf{q} = -\mathbf{J}_{\Phi}^T \mathbf{r}_{\text{tr}}(\mathbf{q}^{(j)}), \quad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta \mathbf{q}.$$

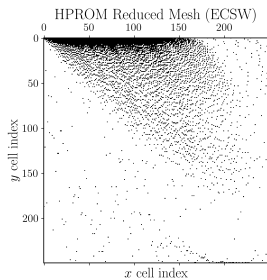
Online Performance (250×250): Max Error over 3 Test Points

Computational model	n (N for HDM)	$\bar{n}$	N_e	$\max \mathbb{RE}_{2,u}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	62 500	–	12.512	–
HPROM	96	–	4 824	1.096	0.738	16.944
HQPROM	39	–	3 505	0.933	0.894	13.992
HPROM-GPR	20	131	1 801	0.845	0.419	29.886
Local HPROM ($N_c = 10$)	14–23	–	1 522	0.956	0.210	59.567
Local HQPROM ($N_c = 10$)	8–12	–	871	0.929	0.287	43.651
Local HPROM-GPR ($N_c = 10$)	6	24–47	541	0.801	0.284	44.103

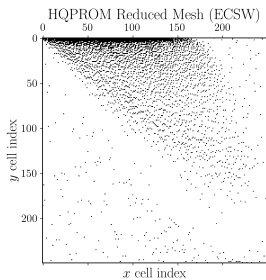
Speedup is computed with respect to the HDM runtime, $t_{\text{HDM}} = 750.702$ s.

N_e denotes the number of sampled ECSW elements ($N_e = 62,500$ for HDM/full mesh).

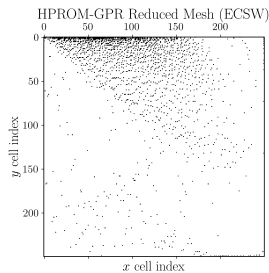
Global Reduced Meshes (ECSW)



HPROM ($N_e = 4,824$)

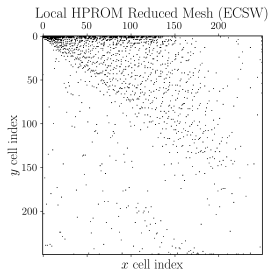


HQPROM ($N_e = 3,505$)

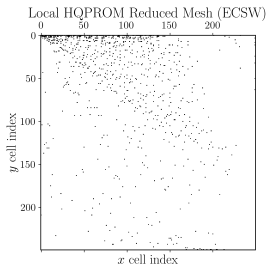


HPROM-GPR ($N_e = 1,801$)

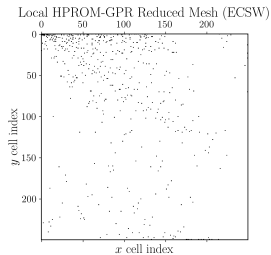
Local Reduced Meshes (ECSW)



Local HPROM ($N_e = 1,522$)

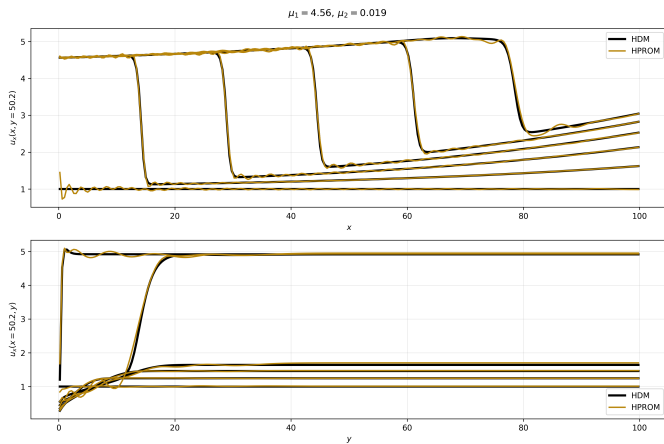


Local HQPROM ($N_e = 871$)

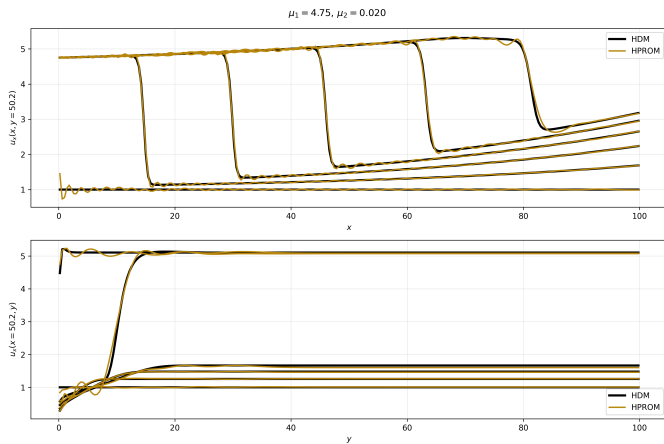


Local HPROM-GPR ($N_e = 541$)

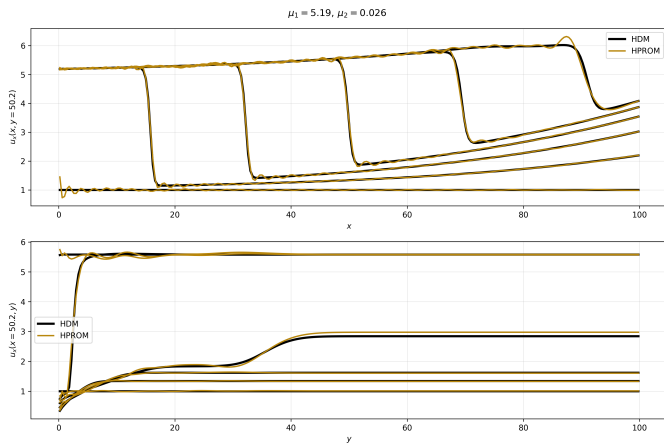
HPROM vs HDM: $\mu = (4.56, 0.019)$



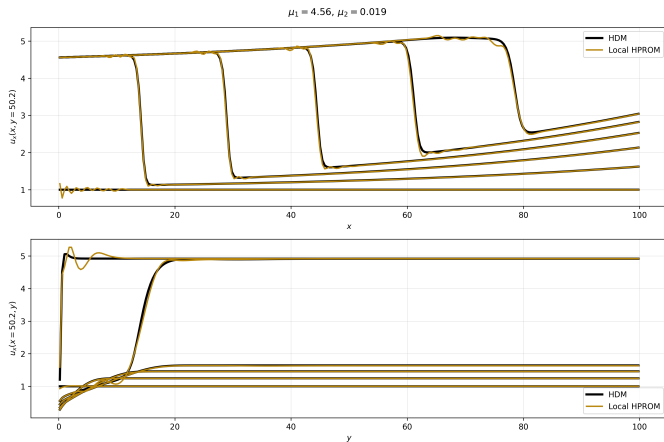
HPROM vs HDM: $\mu = (4.75, 0.020)$



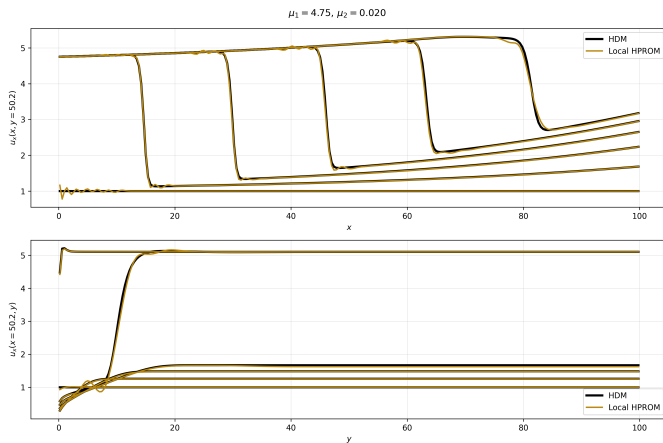
HPROM vs HDM: $\mu = (5.19, 0.026)$



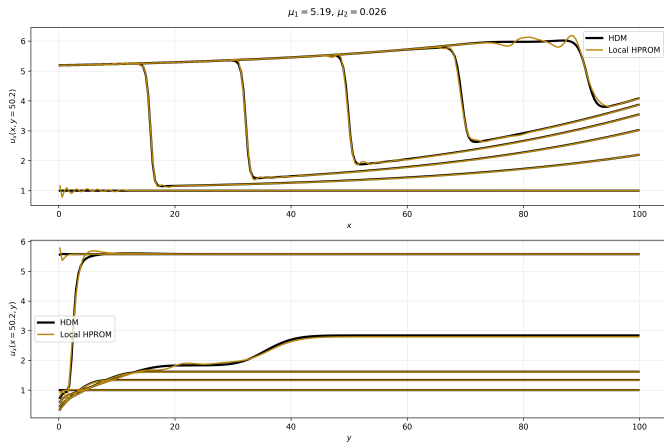
Local HPROM vs HDM: $\mu = (4.56, 0.019)$



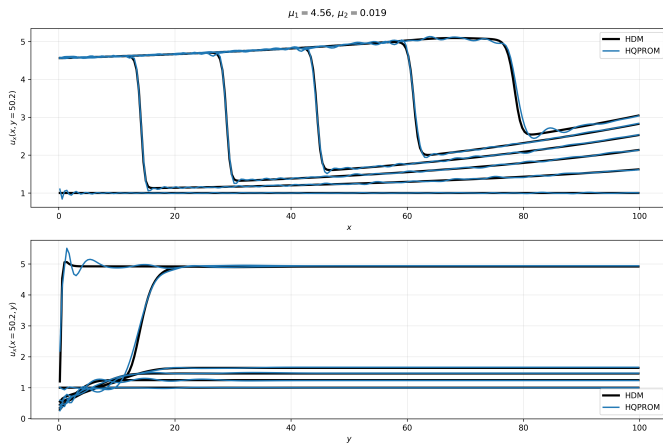
Local HPROM vs HDM: $\mu = (4.75, 0.020)$



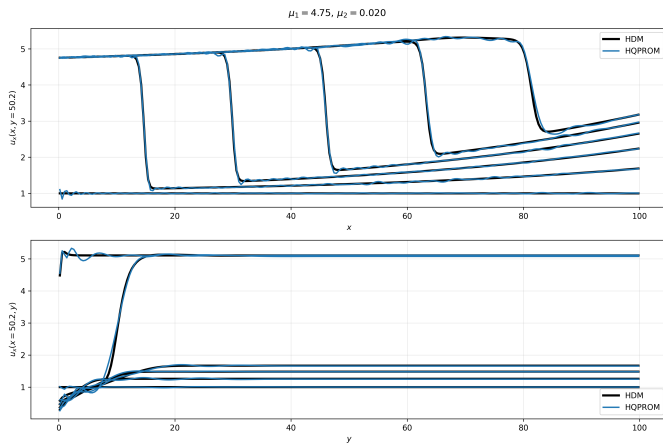
Local HPROM vs HDM: $\mu = (5.19, 0.026)$



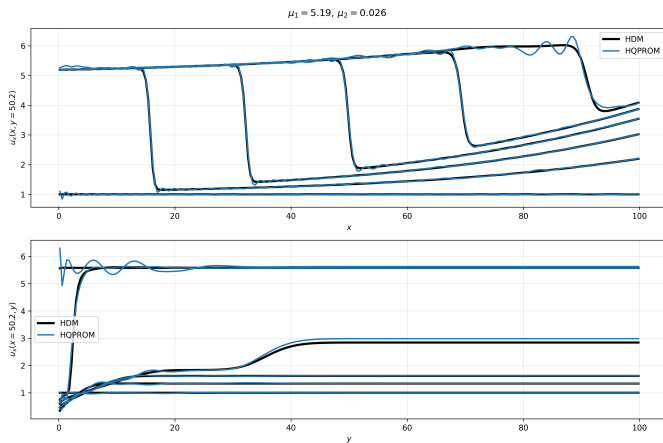
HQPROM vs HDM: $\mu = (4.56, 0.019)$



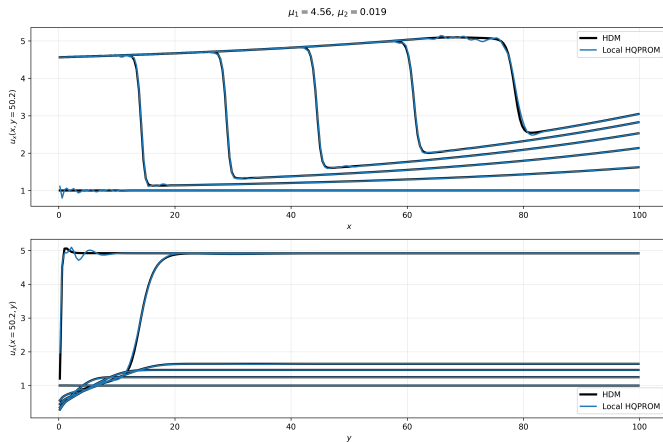
HQPROM vs HDM: $\mu = (4.75, 0.020)$



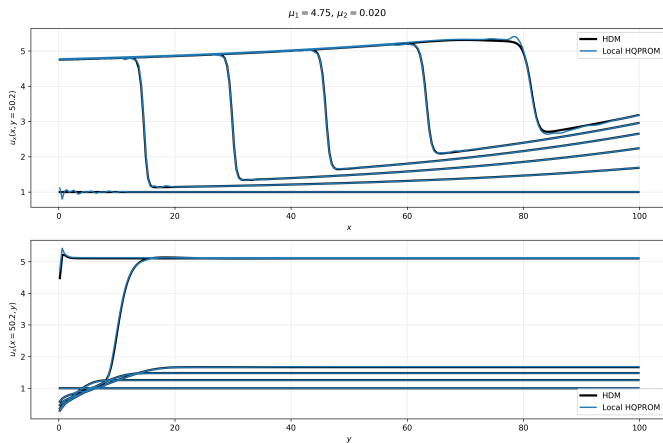
HQPROM vs HDM: $\mu = (5.19, 0.026)$



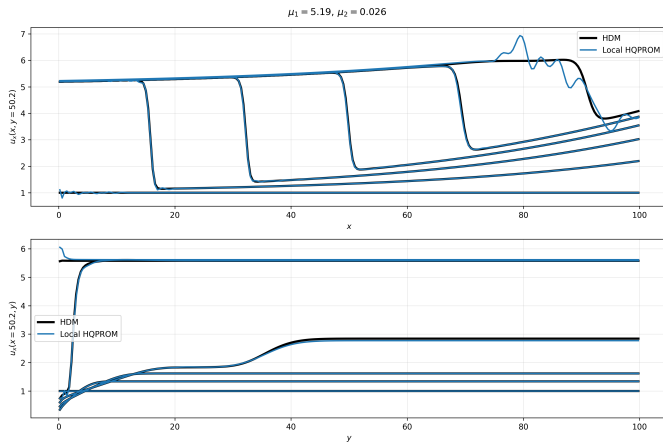
Local HQPROM vs HDM: $\mu = (4.56, 0.019)$



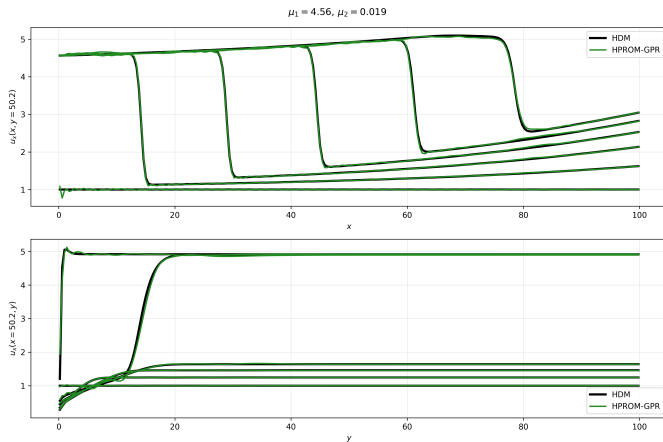
Local HQPROM vs HDM: $\mu = (4.75, 0.020)$



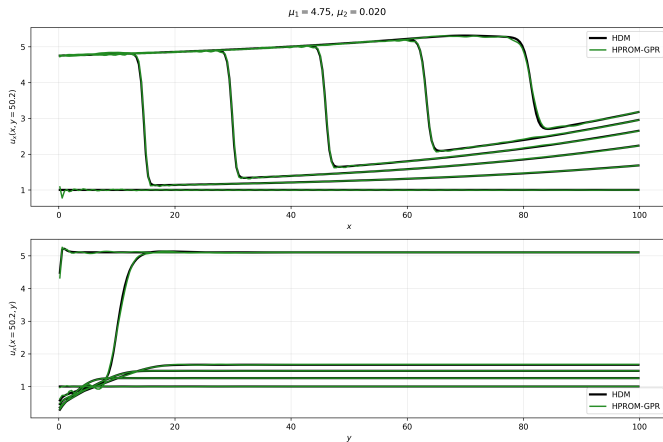
Local HQPROM vs HDM: $\mu = (5.19, 0.026)$



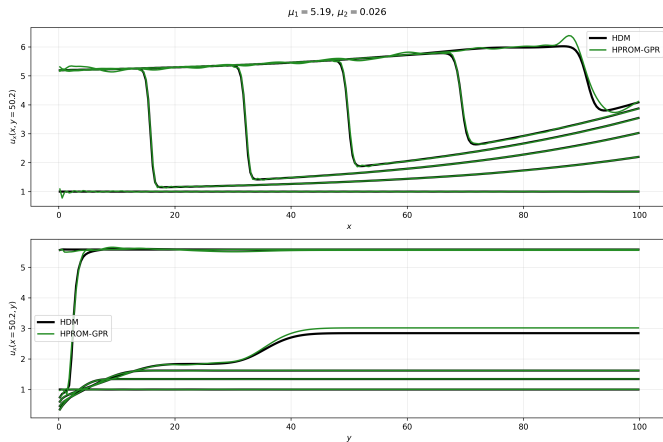
HPROM-GPR vs HDM: $\mu = (4.56, 0.019)$



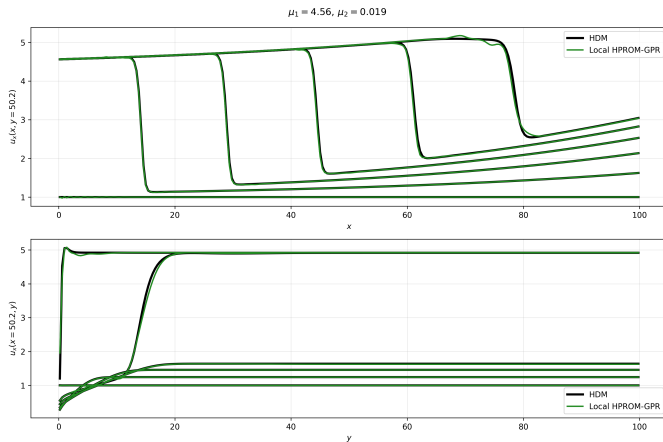
HPROM-GPR vs HDM: $\mu = (4.75, 0.020)$



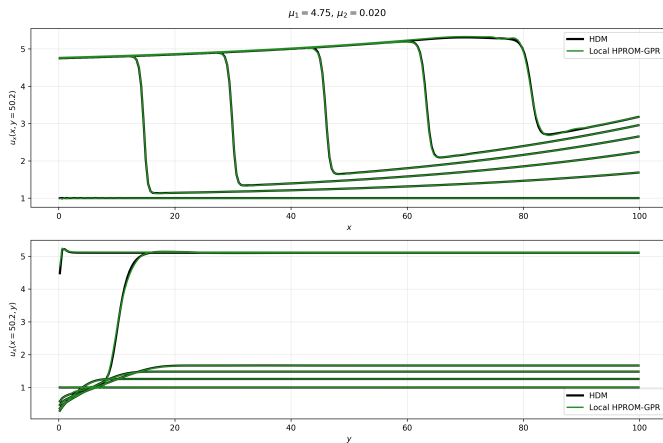
HPROM-GPR vs HDM: $\mu = (5.19, 0.026)$



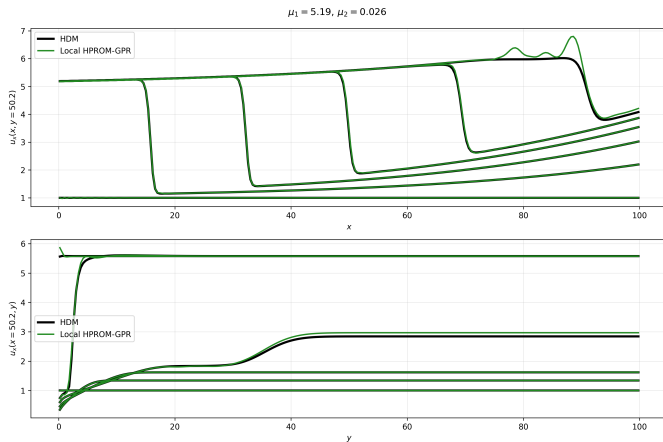
Local HPROM-GPR vs HDM: $\mu = (4.56, 0.019)$



Local HPROM-GPR vs HDM: $\mu = (4.75, 0.020)$



Local HPROM-GPR vs HDM: $\mu = (5.19, 0.026)$



Appendix: Linear Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-} \mathbf{q}_{k^-}, \quad k^+ = \arg \min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

$$\text{Let } \mathbf{a}_{\ell} := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k^-} \mathbf{q}_{k^-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_{\ell} + \mathbf{b}\|_2^2 = \|\mathbf{a}_{\ell}\|_2^2 + \|\mathbf{b}\|_2^2 + 2\mathbf{a}_{\ell}^T \mathbf{b}.$$

$$\|\mathbf{b}\|_2^2 = \|\mathbf{V}_{k^-} \mathbf{q}_{k^-}\|_2^2 \text{ is independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[\|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k^-} \mathbf{q}_{k^-} \right].$$

Precompute

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2,$$

which yields

$$k^+ = \arg \min_{\ell} \left(2\mathbf{g}_{k^-, \ell}^T \mathbf{q}_{k^-} + d_{k^-, \ell} \right).$$

Appendix: HQPROM Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-} \mathbf{q}_{k^-} + \mathbf{H}_{k^-} (\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k^-} \mathbf{q}_{k^-}, \quad \mathbf{c} := \mathbf{H}_{k^-} (\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}).$$

$$k^+ = \arg \min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \arg \min_{\ell} \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2.$$

$$\|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2 = \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\mathbf{a}_\ell^T \mathbf{b} + 2\mathbf{a}_\ell^T \mathbf{c} + 2\mathbf{b}^T \mathbf{c}.$$

$$\|\mathbf{b}\|_2^2, \|\mathbf{c}\|_2^2, 2\mathbf{b}^T \mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[\|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 \right.$$

$$\left. + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k^-} \mathbf{q}_{k^-} + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k^-} (\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}) \right].$$

Define

$$\mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k^-} (\mathbf{q} \otimes \mathbf{q}),$$

then

$$k^+ = \arg \min_{\ell} \left(2\mathbf{g}_{k^-, \ell}^T \mathbf{q}_{k^-} + 2\mathbf{m}_{k^-, \ell}^T (\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}) + d_{k^-, \ell} \right).$$

Appendix: HPROM-GPR Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-} \mathbf{q}_{k-} + \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-}, \quad \bar{\mathbf{q}}_{k-} = \mathcal{N}_{\text{GPR},k-}(\mathbf{q}_{k-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k-} \mathbf{q}_{k-}, \quad \mathbf{c} := \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2$$

$$= \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\mathbf{a}_\ell^T \mathbf{b} + 2\mathbf{a}_\ell^T \mathbf{c} + 2\mathbf{b}^T \mathbf{c}.$$

$$\|\mathbf{b}\|_2^2, \|\mathbf{c}\|_2^2, 2\mathbf{b}^T \mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[\|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k-} \mathbf{q}_{k-} + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-} \right].$$

Hence, with

$$\bar{\mathbf{g}}_{k-, \ell} := \bar{\mathbf{V}}_{k-}^T (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}),$$

$$k^+ = \arg \min_{\ell} \left(2\mathbf{g}_{k-, \ell}^T \mathbf{q}_{k-} + 2\bar{\mathbf{g}}_{k-, \ell}^T \bar{\mathbf{q}}_{k-} + d_{k-, \ell} \right).$$